



bright, metallic luster

HEMATITE

Trigonal • H 5–6 • SG 5.26
Widespread and abundant, this iron oxide is mined extensively for iron. Different forms are coloured black, metallic grey, or brownish red.

FERGUSONITE

Tetragonal
H 5½–6½ • SG 4.2–5.8
Fergusonite is the group name for an oxide of many metals, including yttrium, lanthanum, niobium, and cerium.



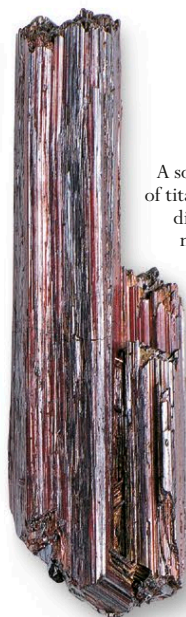
PYROLUSITE

Tetragonal
H 2–6½
SG 5.04–5.08
Pyrolusite is a common manganese oxide. It is the primary ore of manganese, an essential element in steel production.



RUTILE

Tetragonal
H 6–6½ • SG 4.23
A source of titanium, this oxide of titanium often forms impressive displays of thin, translucent needles in quartz crystals.



CHRYSOBERYL

Orthorhombic • H 8½ • SG 3.75
Chrysoberyl is beryllium aluminium oxide. It is a prized gemstone, known for its exceptional hardness and yellow-brown colour.



ZINCITE

Hexagonal • H 4 • SG 5.64–5.68
Zincite is a rare oxide of zinc. Its only significant deposit in the USA has been exhausted.

HYDROXIDES

Hydroxide minerals are compounds of a metallic element and the hydroxyl radical (OH). They are common minerals and often form through a chemical reaction between an existing oxide and fluids rich in water, seeping through Earth's crust. Hydroxides tend to occur in the altered parts of hydrothermal veins and in metamorphic rocks.



GIBBSITE

Monoclinic
H 2½–3 • SG 2.38–2.42
Gibbsite is one of the essential aluminium hydroxides in the aluminium ore bauxite. It also occurs in hydrothermal veins.



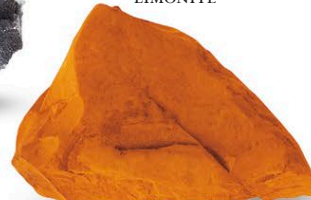
DIASPORE

Orthorhombic • H 6½–7 • SG 3.2–3.5
Diaspore is an aluminium hydroxide which is found in bauxite. It also occurs in marble and altered igneous rocks.



BOTRYOIDAL
GOETHITE

LIMONITE



GOETHITE

Orthorhombic • H 5–5½ • SG 4.27–4.29
Goethite is a common iron hydroxide. A non-crystalline hydrated variety – limonite – gives a yellow-brown colour to soils and rocks.



BRUCITE

Trigonal • H 2½–3 • SG 2.39
Brucite is a magnesium hydroxide, and its colour is white, grey, blue, or green. It occurs in metamorphic rocks.



STIBICONITE

Cubic • H 5½–7 • SG 3.5–5.5
An uncommon hydroxide of antimony, stibiconite is white or yellowish brown, and forms by the alteration of other antimony minerals, especially stibnite.



LEPIDOCROCITE

Orthorhombic • H 5
SG 4.05–4.13
This relatively rare iron hydroxide may occur with goethite. It is reddish brown and can form irregular and fibrous shapes.



ROMANÈCHITE

Monoclinic • H 5–6 • SG 6.45
Dark and opaque, this oxide of manganese with barium is usually found in aggregates or in massive forms. Crystals are rare.



BAUXITE

Amorphous mixture
H 1–3 • SG 2.3–2.7
The primary aluminium ore, bauxite is not a single mineral, but rather an assemblage of aluminium hydroxides and iron oxides.